

Experiment Matrix and Result-Table Blueprint

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Abstract

This document fixes the paper-facing experiment tables before confirmation runs. The overview binds each research question to a track and result table; the three result tables bind policy rows to primary endpoints, service measures, and mechanism interventions. Numeric cells remain blank by design. They will be populated only from frozen experiment cards, raw event manifests, and stream-level uncertainty calculations.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Cooperation and coordination*.

Keywords

multi-agent systems, role formation, peer judgment, online adaptation, experiments

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1 Table set and correspondence

The paper should use one compact overview and three result-bearing tables. The overview is a routing table, not an empirical result. Table 2 is the hero comparison for the primary ArtifactRole track and answers RQ1–RQ2. Table 3 reports the service and safety endpoints for RQ3–RQ4 across the two tracks. Table 4 isolates the mechanism and responsibility boundaries. The exact nine-cell execution manifest remains the source of roots, seeds, budgets, scorers, stopping rules, and raw-log digests.

Placeholder rule. A dash is an unfilled empirical cell, never zero and never a predicted value. Each result table will report stream-level estimates with the interval method fixed in the card. Counts, costs, and failures remain visible in the artifact manifest even when a primary cell is UNKNOWN.

2 Primary result tables

2.1 ArtifactRole: information and assignment

The first result table is the hero table. Its policy rows are the complete matched-information comparison boundary, while its columns are limited to the two primary scientific questions and complete cost. The row order is fixed before confirmation; the last row is the candidate mechanism, not a claim of superiority.

2.2 Online service and stability

The second result table keeps operational claims separate from the hero utility comparison. PeerSelect supplies the controlled local-stream service measurement; ArtifactRole supplies the responsibility mutations. A method cannot trade away attribution safety for a lower update latency without showing that trade-off here.

2.3 Mechanism and responsibility ablations

The third table answers why any observed gain occurs. It holds the task roots, menus, models, and complete-cost accounting fixed while removing one information or update component at a time. These rows are not additional baselines; they are interventions on the candidate mechanism.

3 Population and result-entry contract

The tables will be populated in a fixed order: first the independent-root and same-information checks, then the primary ArtifactRole result, then the PeerSelect service/stability table, and finally the ablation table. A table row is reported as UNKNOWN when its scorer, transport, or responsibility contract cannot observe the required quantity. Failed attempts retain their measured cost; they are not deleted to improve the mean. The final paper will add stream-level intervals, per-root variation, and a short error analysis next to the relevant table rather than hiding these quantities in a single aggregate number.

This blueprint is an internal pre-results artifact. It fixes presentation and estimand alignment; it does not assert that any arm wins or that the benchmark and baseline gates have passed.



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RQ	Track / unit	Policy contrast	Primary result table	Primary estimand
RQ1	ArtifactRole / independent root	RARE vs raw acceptance, terminal-only, and contextual-trust-linear	Table 2	Held-out Brier/log loss and calibration
RQ2	ArtifactRole / target assignment	Evidence plus delayed credit vs no evidence, delayed-only, and pooled	Tables 2 and 4	Future quality–cost utility and assignment change
RQ3	PeerSelect / independent stream	Updater vs uniform, no-history, and history-only	Table 3	Payoff/regret, update p95, backlog, state bytes
RQ4	Both / mutation and drift streams	Ownership mutation, correction/replay, and leave-one-out	Tables 3 and 4	False attribution, UNKNOWN precision, forgetting/recovery

Table 1: Paper-facing experiment map. Each row points to the table that will carry the corresponding measured contrast. The table set is an overview of the pre-registered matrix; it is not evidence until the matching confirmation card is executed.

Policy	RQ1: situated signal		RQ2: future assignment		Cost	Streams
	Brier ↓	Calibration ↓	Quality–cost utility ↑	Assignment change ↑	Complete cost ↓	
Uniform	—	—	—	—	—	—
No update	—	—	—	—	—	—
Raw acceptance	—	—	—	—	—	—
Terminal only	—	—	—	—	—	—
Contextual-trust-linear	—	—	—	—	—	—
Pooled controller	—	—	—	—	—	—
RARE	—	—	—	—	—	—

Table 2: Primary ArtifactRole result table. Every arm receives the same candidate menu, opportunity schedule, public-information boundary, selected-only rule, and complete-cost contract. Values are filled only after the independent root, later assignment, and outcome checks pass; intervals are stream-level intervals from the frozen card.

Policy / intervention	Track	RQ3: service			RQ4: safety and stability		
		Update p95 ↓	Backlog p95 ↓	State bytes ↓	False attribution ↓	UNKNOWN precision ↑	Forgetting / recovery ↓
No update	PeerSelect	—	—	—	—	—	—
History only	PeerSelect	—	—	—	—	—	—
Contextual-trust-linear	PeerSelect	—	—	—	—	—	—
RARE	PeerSelect	—	—	—	—	—	—
Recipient-only mutation	ArtifactRole	—	—	—	—	—	—
Mixed ownership mutation	ArtifactRole	—	—	—	—	—	—
Delayed / out-of-order feed-back	Both	—	—	—	—	—	—

Table 3: Service and stability table. Latency, backlog, and state size are measured under a fixed arrival and resource budget. Mutation rows test whether recipient-owned, mixed, delayed, or out-of-order events produce safe no-op or UNKNOWN behavior rather than false producer credit.

Intervention	Removed or changed component	Primary endpoints			
		Future utility ↑	Assignment change ↑	False attribution ↓	Complete cost ↓
No evidence / no update	Public evidence and persistent update	—	—	—	—
Public evidence only	Persistent delayed update	—	—	—	—
Delayed update only	New public evidence	—	—	—	—
Evidence + delayed update (RARE)	None	—	—	—	—
RARE without ownership gate	Structural attribution gate	—	—	—	—
RARE without selected-only rule	Selected-only credit restriction	—	—	—	—
RARE without correction/replay	Version correction and replay checks	—	—	—	—

Table 4: Mechanism and responsibility ablations. The primary interpretation requires both a change in future assignment and a valid independent outcome. A utility gain without attribution safety, or a safety gain without changed assignment, is reported as a boundary result rather than role formation.